



IOTRICITY S3

Team Name

ChargeX

Problem Statement

Homes, farms, industries rely on manual tank checks, leading to overflow, water wastage, and pump damage from dry runs due to no real-time visibility



Problem Identification & Analysis

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What is the real-world problem?

- Water tanks overflow or run dry unnoticed
- Manual checking wastes time and water
- Motor left ON/OFF by guesswork burns pumps



Who faces it, and where?

- Households, apartments and hostels
- Farmers and irrigation tanks
- Schools, factories and small industries



How is it solved today?

- Visual checks and manual switching
- Float switches / basic mechanical valves
- Costly commercial automation systems



Gaps in existing solutions

- No remote view of the water level
- No low-water or overflow alerts
- Expensive, hard to install or customize

Identified Problem

“Water tanks are monitored and refilled manually, causing overflow, dry-running pumps and wasted water, with no remote visibility or alerts.”

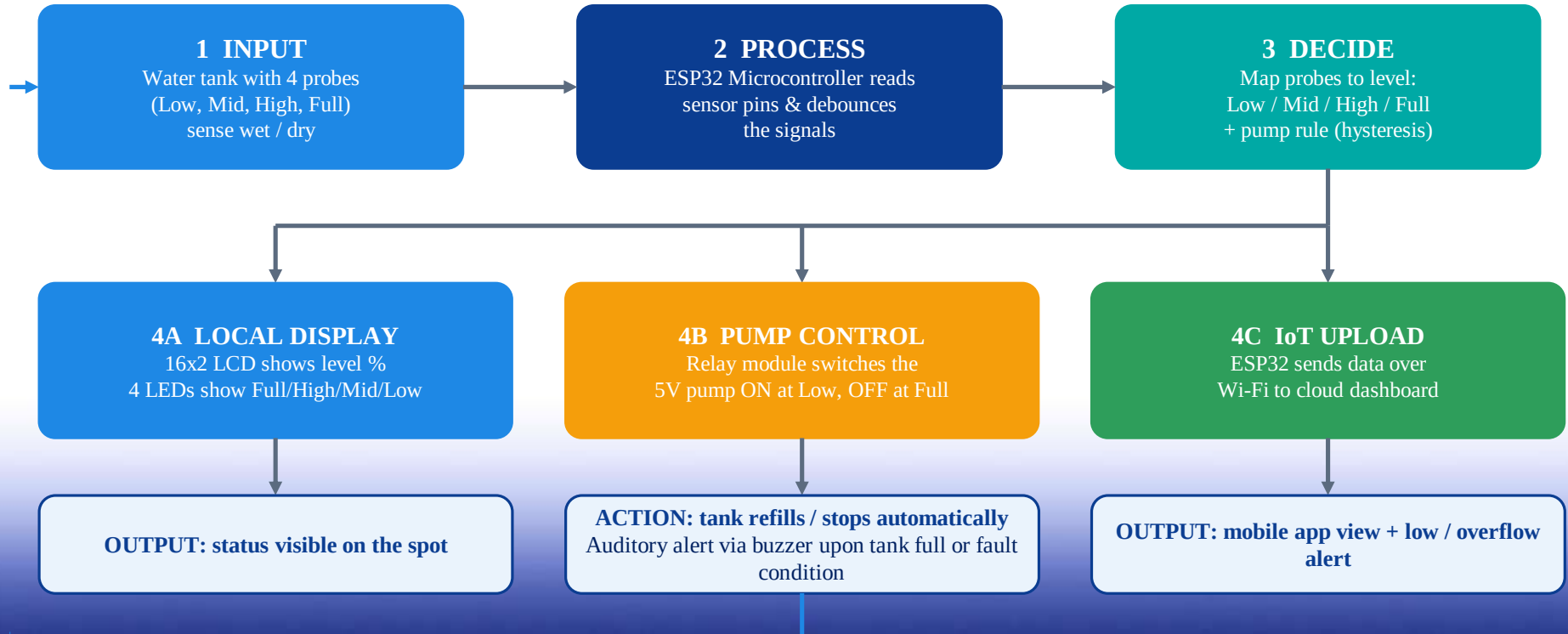
Our Proposed Solution

“We propose to develop an IoT-based water level monitoring and automatic pump control system that will prevent overflow and dry-run, save water, and send real-time alerts to a mobile app.”



Proposed Technical Approach — Working Flow

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Feedback loop: rising water level is sensed again by the probes— the cycle repeats continuously



Technology Stack

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HARDWARE



Water Level Ultrasonic Sensors

4 probes (Top, High, Mid, Low) detect water at each level



Relay Module

Safely switches the pump ON/OFF from low-voltage signals



LED Indicators

Red / Orange / Yellow / Green tank status



ESP32 MICROCONTROLLER & Buzzer (audio alert)

Microcontroller: reads sensors, runs logic, drives outputs, Buzzer triggers sound alert for full tank detection.



Water Pump (5V)

Fills the tank automatically



5V DC Supply

Stable power for the circuit and pump



ESP32 Wi-Fi

Sends level data to cloud and mobile app



16x2 LCD

Shows live level and status locally



Tank, Board, Wires

Transparent tank, base board, jumpers, tubing

SOFTWARE



Arduino IDE + C/C++

Write and upload the control code



Libraries

LiquidCrystal_I2C, ESP32WiFi, IoT platform library



IoT Cloud Platform

e.g. Blynk / ThingSpeak stores and streams data



Mobile App / Dashboard

Live level, status, history graph, alerts



Feasibility & Viability

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Technical Feasibility

- Uses common, proven modules
- Simple digital I/O + Wi-Fi upload
- Buildable with basic Arduino skills



Resources Required

- Components listed in Tech Stack
- Laptop, Arduino IDE, Wi-Fi hotspot
- Multimeter, soldering / breadboard



Budget (approx.)

- ESP32 Microcontroller ~₹300
- Pump, relay, LCD, probes ~₹720
- Tank, supply, wires, LEDs ~₹500
- Buzzer/Alarm module ~₹50
- Total: about ₹1000–1600



Time

- Wk 1: design & procure parts
- Wk 2: wiring & coding
- Wk 3: IoT app & testing
- Fits a 3–4 week timeline



Challenges → Fixes

- Probe corrosion → use SS probes
- Wi-Fi drops → local LCD/LED backup
- Pump dry-run → low-level cutoff



Future Scope

- Easy to maintain: low-cost parts
- Modular: add sensors or tanks
- Scales to homes, farms, industry



Impact & Benefits

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Homes & Apartments



Farmers



Industry & Factories



Schools & Communities



Key Benefits

- Time: no manual tank checks
- Cost: avoids wasted water & pump wear
- Safety: prevents overflow damage
- Convenience: monitor from anywhere



Expected Impact

- Less water wastage
- Longer pump life
- Smarter, data-driven water use
- Low-cost model others can copy



Impact Measurement

- Overflow / dry-run events before vs after
- Water & electricity use per week
- Alert response time in app
- User feedback from test households



Limitations & Future

- Limits: needs Wi-Fi; probes need cleaning
- Add ultrasonic/flow sensors, solar power
- Add SMS alerts and usage prediction



Level Logic: Sensors → Action

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Tank State	Probes Wet	LED	Pump	App / Alert
Empty	None	Red blinks	ON	Critical low alert
Low	Low	Red	ON	Low water alert
Mid	Low, Mid	yellow	ON (filling)	Status: Normal
High	Low, Mid, High	Green	ON (filling)	Status: Normal
Full	All four	Blue	OFF	Tank full / overflow alert

Control Logic (pseudo-code)

```
if level == LOW → pump ON
if level == FULL → pump OFF
update LCD, LEDs, then upload via ESP32
```

Why hysteresis?

The pump starts at Low and stops only at Full, so it does not rapidly switch ON/OFF when the water ripples. This protects the relay and pump.



Roadmap & Conclusion

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Phase 1

Prototype

4-level sensing, LCD, LEDs,
auto pump

Phase 2

IoT Live

Cloud dashboard, mobile app,
alerts

Phase 3

Upgrade

Ultrasonic + flow sensor, solar
power, SMS

Phase 4

Scale

Multi-tank, usage analytics,
prediction

- **Wokwi simulation(Circuit+Code):**<https://wokwi.com/projects/476309690117978113>
- **GitHub Repository:**<https://github.com/Kriti2706/Smart-water-managements.git>

Conclusion

A low-cost, IoT-enabled system that senses the water level, controls the pump automatically and alerts users in real time — saving water, time and money for homes, farms and industry.

Save Water | Smart Monitoring | Better Tomorrow Thank You!